Owusu Kantankah

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EDUCATION

University of Wisconsin-Madison **Madison, WI**
Bachelor of Science in Computer Science and Data Science
Cumulative GPA: 3.43
Relevant Coursework: High Performance Computing (Graduate), Computer Graphics, Algorithms, Data Structures & Algorithms III, Database Management Systems, Linear Algebra.
Affiliations: Point72 Academy Spring Sessions, Goldman Sachs Possibility Series, Colorstack, NSBE, CodePath TIP 102

PROFESSIONAL EXPERIENCE

LASER Research Program, UW-Madison **Madison, WI**
Undergraduate Machine Learning Researcher *Summer 2026*

- Engineered a MARL training pipeline for 250 simulated drone agents, ingesting structured flight telemetry to study emergent flocking behavior, benchmarked against the classical Boids algorithm.
- Implemented REINFORCE and A2C policy gradient algorithms from scratch in PyTorch, producing a learned decision boundary visualized across the full state space.
- Selected to present findings at UW-Madison's Undergraduate Research Symposium.

Space Science and Engineering Center **Madison, WI**
Undergraduate Data Engineering Researcher *September 2025 - May 2026*

- Built Python data pipeline (NumPy, pandas) processing **10GB+** of NetCDF spacecraft telemetry from JAXA's Akatsuki orbiter, cutting analysis runtime by **35%** and enabling faster iteration.
- Identified a **~195K** polar vortex and **20K** equator-to-pole thermal gradient via zonal-mean analysis on 1440×720 NetCDF grids, contributing to ongoing research on Venus cloud-top absorbers; **presented at UW-Madison's Undergraduate Research Symposium.**

StudioNil **Remote**
Founder & Lead Software Developer *March 2023 - August 2025*

- Founded and shipped a multiplayer game that reached **10,000** MAU in 6 weeks, architected client-server infrastructure handling **100** concurrent players per instance at **0.02%** crash rate.
- Engineered a custom 3D spatial hitbox system using raycasting and geometric detection, replacing Roblox's native .Touched API and improving collision accuracy by **1.8x** in real-time combat.

PROJECTS

CPU Raytracer [Link] - *C++, GCC, CMake, Git* *June 2026*

- Built a CPU-based ray tracer in C++ implementing ray-sphere intersection, diffuse/metal/dielectric materials, reflections, refractions, antialiasing, and depth-of-field camera effects.

3D Drone Simulation [Link] - *Three.js, GLSL, Javascript, HTML, CSS, Git* *May 2026*

- Built a 3D drone swarm simulation in Three.js with custom GLSL shaders, implementing Reynolds' boids algorithm (separation, cohesion, alignment) to drive emergent flocking across **1000** agents.
- Collapsed **1000** per-drone draw calls into a single InstancedMesh dispatch with per-instance transform matrices, improving FPS from **80** to **133,** and halving GPU memory to **9 MB**.

Centralized Morgridge Event Calendar [Link] - *Python, Flask, SQLite, AWS, Git* *February 2026*

- Built RESTful Flask API with SQLite database supporting event retrieval, insertion, and channel management, deployed on AWS EC2 as the data layer between automated scrapers and a React frontend; contributed as backend engineer on a **9**-person team.

SKILLS

- **Languages**: Java, Python, C++, C, GLSL, SQL, TypeScript/JavaScript, HTML/CSS
- **Frameworks & Libraries**: PyTorch, NumPy, pandas, Matplotlib, Three.js, React, Flask, Tailwind
- **Developer Tools:** Git, GitHub, Linux, AWS (EC2), SQLite, CMake, Visual Studio Code